PHYS 4041/5041:

Mathematical Methods of Physics 1

Series

Fall 2023

OHIO UNIVERSITY

Department of physics + astronomy OHIO UNIVERSITY

Series

- Finite and infinte series are very important mathematics and physics.
- The solution of many problems in physics results in a series (e.g., differential equations in this course, perturbation theory in quantum mechanics).
- Series may be summed using a computer. Most special functions, such as $\exp(x)$, are computed this way.

Sequences

- A sequence is a discrete list of terms, e.g., $S_n = n/(n+1)$ for $n = 0, 1, 2, \ldots$.
- One is often interested in the $n \to \infty$ limit of the sequence, $\lim_{n\to\infty} S_n$. What is this limit for our example?

Series

- A series (or sum) is a sequence defined according to

$$S_n = \sum_{i=0}^{n} a_i.$$

- If $n \to \infty$, we call this an infinite series. We are particularly interested in this case.

- Example, *geometric series*: $a_n = ar^n$, such that $S_n = a + ar + ar^2 + \ldots ar^n$.

- In this case, $S_n = a(1 - r^{n+1})/(1 - r)$. Consider what happens if $n \to \infty$. For what values of r does this series converge?

- Going forward, we will assume that our series are infinite, unless otherwise indicated.

Convergence

- It is critical to be able to establish whether *or not* an infinite series converges.
- A *necessary condition* is that $\lim_{n\to\infty} a_n = 0$. This is the *preliminary test*.
- The *comparison test*:
  If $\sum_{n=0}^{\infty} b_n$ converges, and $0 \leq a_n \leq b_n$, the series $\sum_{n=0}^{\infty} a_n$ also converges.
  Conversely, if $a_n \geq b_n \geq 0$ and $\sum_{n=0}^{\infty} b_n$ diverges, then $\sum_{n=0}^{\infty} a_n$ also diverges.

Ratio Test

- In practice, the *ratio test* is often the most useful tool.
- Define

$$\rho_n = \left| \frac{a_{n+1}}{a_n} \right|,$$
$$\rho = \lim_{n \to \infty} \rho_n.$$

- Then

$$\begin{aligned} \rho < 1 &\quad \text{the series converges,} \\ \rho = 1 &\quad \text{indeterminent} \\ \rho > 1 &\quad \text{the series diverges.} \end{aligned}$$

- Note the similarity to the geometric series here.

Other Tests

- There are many additional tests available.
- The *integral test* involves a comparison to an integral that bounds the sum. It has a nice graphical interpretation. See Boas, section 1.6.
- The conditions for convegence for an alternating series are considerably relaxed. If the signs of a_n alternate, $|a_{n+1}| \leq |a_n|$, and $\lim_{n\to\infty} a_n = 0$, then the series converges.
- If $\sum_{n=0}^{\infty} |a_n|$ converges, the series is said to be *absolutely convergent*.
- If a sum converges, but is *not* absolutely convergent, it is said to be *conditionally convergent*.

Algebra with series

- Convergent series may be added, subtracted, or multiplied by a constant on a term-by-term basis.
- The terms in an absolutely convergent series may be re-ordered without changing the value of the series. This is *not* the case for conditionally-convergent series.
- For example, two absolutely convergent series may be multiplied to yield another absolutely convergent series:

$$S_1 = \sum_{i=0}^{\infty} a_i, \qquad S_1 = \sum_{i=0}^{\infty} b_i$$

$$S_1 S_2 = \sum_{i=0}^{\infty} c_i$$

$$c_n = \sum_{i=0}^{n} a_i b_{n-i}.$$

Power series

- A *power series* is defined to be

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \ldots,$$

  where x is an independent variable. More generally, it is defined in powers of $(x - a)$, where a is constant.

- The ratio test yields $\rho = \lim_{n\to\infty} |a_{n+1}x/a_n|$. The series thus converges if $|x| < \lim_{n\to\infty} |a_n/a_{n+1}|$.

- This inequality defines the *interval of convergence*. If x is complex, this generalizes to the *radius of convergence*.

- Also note that if $|x| > \lim_{n\to\infty} |a_n/a_{n+1}|$, the series diverges. On the edges of the interval, the result is indeterminant (you may find conditional convergence here).

Facts about power series

- Power series may be differentiated or integrated with respect to the independent variable term by term, without changing the radius of convergence.
- Power series may be added, subtracted, multiplied, or divided.
- One power series may be substituted into another power series.
- The power series of the form $\sum_{n=0}^{\infty} a_n(x-a)^n$ is unique, provided that x is in the interval of convergence.

Taylor series

- The Taylor series expansion of $f(x)$ around $x = a$ is given by

$$f(x) = \sum_{n=0}^{\infty} \frac{(x-a)^n}{n!} f^{(n)}(a)$$

$$= f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2} f''(a) + \frac{(x-a)^3}{3!} f^{(3)}(a) + \ldots$$

- Often, we have $a = 0$. This case may be called the Maclaurin series.

- Repeated differentiation is one way to determine the power series coefficients. It quickly becomes tedious if you need higher-order terms and $f(x)$ is complicated.

- In complicated cases, it is usually easier to use "algebra with power series." Consider $\cos x/[1 + \exp(x)]$.

Examples for consideration

Consider the Taylor series expansions of these functions around $x = 0$:

- $\log(1 + x)$
- $\dfrac{\log(1 + x)}{x}$
- $\dfrac{1}{1 - \exp(x)}$
- $(1 + x)^3$
- $\sin x \, \cos x$

Binomial theorem

- The Binomial theorem is a particularly important case:

$$(1+x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \dots$$

- It terminates if p is a non-negative integer. It is a geometric series if $p = -1$.

- This seems like a good time to mention that the formula for the coefficients a_n is sometimes very simple when defined recursively. In this case, we have $a_0 = 1$ and for the higher orders

$$a_{n+1} = \frac{p-n}{n+1} a_n.$$

  This is how you would do the calculation on a computer.

Applications of Taylor series

(Why should I care?)

- L'Hôpital's rule can be derived from

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f(a)+(x-a)f'(a)+\dots}{g(a)+(x-a)g'(a)+\dots},$$

  which is useful when when $f(a)=g(a)=0$.

- Approximating integrals or other expressions that you could not otherwise do.

- Solving differential equations.

Another application of Taylor series

- Limiting cases of a formula or theory may provide physical insight.
- Example: The energy of a particle in special relativity is related to its mass m and speed v via

$$E = \frac{mc^2}{\sqrt{1 - \left(\frac{v}{c}\right)^2}},$$

where c is the speed of light.

- Using the Taylor series of a binomial, we find

$$\begin{aligned} E &= mc^2 \left[1 + \frac{1}{2}\left(\frac{v}{c}\right)^2 + \ldots\right] \\ &= mc^2 + \frac{1}{2}mv^2 + O\left[\left(\frac{v}{c}\right)^4\right], \end{aligned}$$

which can be interpreted as the rest-mass energy + non-relativistic kinetic energy + relativistic corrections.